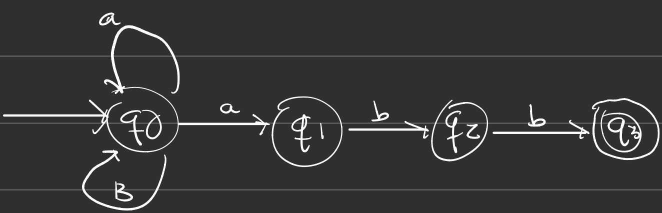
1)



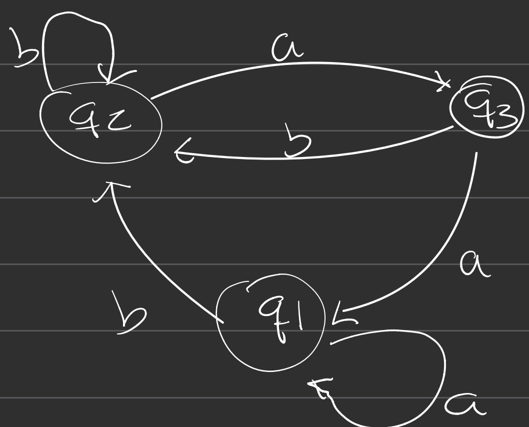
$$q_0 = \varepsilon + q_0 a + q_0 b = \overset{Q}{\varepsilon} + \overset{R}{q_0}(\overset{P}{a+b}) = \overset{Q}{\varepsilon}(\overset{P^*}{a+b})^* = (a+b)^*$$

$$q_1 = q_0 a = (a+b)^* a$$

$$q_2 = q_1 b = (a+b)^* ab \qquad R_f = (a|b)^* abb$$

$$q_3 = q_2 b = (a+b)^* abb$$

2)



$$q_1 = \varepsilon + q_1 a + q_3 a =$$

$$q_2 = q_2 b + q_1 b + q_3 b$$

$$q_3 = q_2 a$$

$$q_1 = \varepsilon + q_1 a + q_2 a^2$$

$$q_2 = q_2 b + q_1 b + q_2 ab = \overset{R}{q_2}(\overset{P}{ab+b}) + \overset{Q}{q_1 b}$$

$$q_2 = q_1 b (ab+b)^*$$

$$q_1 = \varepsilon + q_1 a + q_1 b (ab+b)^*$$

$$q_1 = \varepsilon + q_1 (a + b + (ab+b)^*)$$

$$q_3 = b(q_1 b (ab+b)^*) = q_1 b^2 (ab+b)^*$$

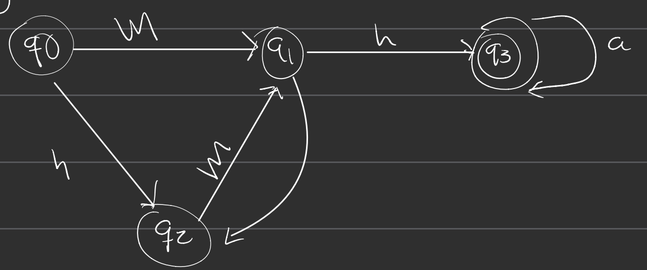
$$q_1 = (a+b+(ab+b)^*)^* = (a+b)^*$$

$$q_2 = (a+b)^* b (a+b)^*$$

$$q_3 = (a+b)^* b^2 (ab+b)^*$$

$$RF = (a|b)^* (ab|b)^* b^2$$

3)



$q_0 = \varepsilon$ $\qquad$ $q_2 = q_1 a + q_0 h$

$q_1 = q_0 m + q_2 m$ $\qquad$ $q_3 = q_3 a + q_1 h$

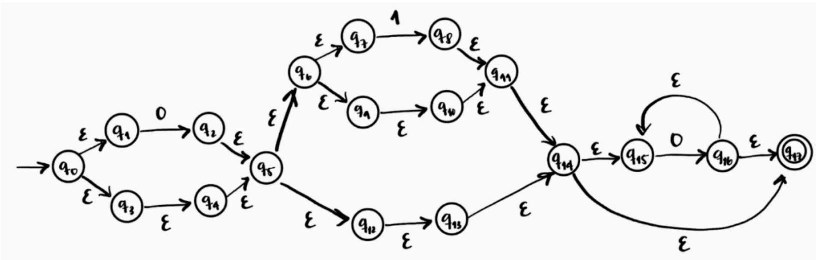
$$q_1 = m + (q_1 a + h) m = m + q_1 am + hm = q_1(am) + (m+hm)$$

$$q_1 = (m+hm)(am)^*$$

$q_3 = q_3\, a + a_1 h = a_1 h\, a^* = (m + hm)(am)^* h a^*$

Problema 2: 25%

Utilice el Lema de Arden para encontrar la expresión regular del siguiente autómata. Deje todo su procedimiento



$q_0 = \varepsilon$

$q_2 = q_0$

$q_3 = q_1 0$

$q_4 = q_3$

$q_5 = q_2 + q_4$

$q_6 = q_5$

$q_7 = q_6$

$q_8 = q_7 1$

$q_9 = q_6$

$q_{10} = q_9$

$q_{11} = q_8 + q_{10}$

$q_{12} = q_5$

$q_{13} = q_{12}$

$q_{14} = q_{11} + q_{13}$

$q_{15} = q_{14} + q_{16}$

$q_{16} = q_{15} 0$

$q_{17} = q_{14} + q_{16}$

$q_5 = q_2 + q_4 = 0 + \varepsilon = (\varepsilon|0)$

$q_6 = q_5 = (\varepsilon|0)$, $q_7 = q_6 = (\varepsilon|0)$

$q_8 = q_7 1 = (\varepsilon|0)1$

$q_9 = q_6 = (\varepsilon|0)$, $q_{10} = q_9 = (\varepsilon|0)$

$q_{11} = q_8 + q_{10} = (\varepsilon|0)1 + (\varepsilon|0) = (\varepsilon|0)(1|\varepsilon)$

$q_{12} = q_5 = (\varepsilon|0)$, $q_{13} = q_{12} = (\varepsilon|0)$

$q_{14} = q_{11} + q_{13} = (\varepsilon|0)(1|\varepsilon) + (\varepsilon|0) = (\varepsilon|0) = (\varepsilon|0)((1|\varepsilon) + \varepsilon) =$

$= (\varepsilon|0)(1|\varepsilon)$

$q_{15} = q_{14} + q_{16}$ $\quad q_{16} = q_{15}0$

$q_{15} = q_{14} + q_{15}0$ $\quad q_{16} = q_{15}0 = q_{14}00^*$

$q_{17} = q_{14} + q_{16} = q_{14} + q_{14}00^* = q_{14}(\varepsilon + 00^*) = q_{14}0^*$

$q_{17} = q_{14}0^* = (\varepsilon|0)(1|\varepsilon)0^*$ $\quad R_F = (\varepsilon|0)(1|\varepsilon)0^*$